EcoKitchen AI - Project Proposal



Food Waste Management and Donation Platform

An Intelligent Food Waste Management and Donation Platform Using Artificial Intelligence

Prepared By :
Group White Builders (Group 29)

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Technology Challenge Competition – Project Proposal

01. Software Service or Product Development

Product Development	Survey Details: A preliminary survey and observations conducted among restaurants, hotels, supermarkets and food donation organizations revealed that significant edible food is wasted daily due to inaccurate demand estimation, poor inventory managementand lack of an efficient donation mechanism. The survey indicated a need for an AI-powered solution to predict food demand and redistribute surplus food to nearby charities.

02. Project Title

EcoKitchen AI: An Intelligent Food Waste Management and Donation Platform Using Artificial Intelligence

03. Project Background

Food waste is a major environmental and social challenge. Restaurants, hotels, supermarkets and catering services often prepare more food than required because of inaccurate demand forecasting. Large quantities of edible food are discarded while many people experience food insecurity. EcoKitchen AI analyzes historical sales data to predict daily food demand, helping businesses prepare optimal quantities. The system identifies surplus food, sends expiry alerts and connects businesses with nearby charities for donations, promoting sustainability and reducing operational costs.

04. Project Type

☑ **Software-Based**

☐ Hardware-Based

☐ Both

05. Group Details

Role	Name	Student ID	Email	Contact No.
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06. Problem Statement

Food businesses generate substantial food waste due to inaccurate demand forecasting, inefficient inventory management and limited mechanisms for redistributing surplus food. An AI-powered platform is needed to forecast demand, reduce waste, monitor inventory and facilitate food donations.

07. Project Objectives

1. Develop an AI-based food demand prediction system.
2. Reduce food waste through inventory recommendations.
3. Connect businesses with nearby charities.
4. Generate reports for decision making.

08. Project Description

EcoKitchen AI is a web platform that collects sales and inventory data, predicts food demand using machine learning, tracks inventory and expiry dates, records food waste and enables surplus food donation.

Main Features:
• Restaurant Dashboard
• AI Food Demand Prediction
• Inventory Management
• Food Waste Tracking
• Donation Management
• Expiry Alerts
• Charity Location Mapping
• Smart Notifications

09. Methodology

Development Approach

The project will follow the **Agile Software Development Methodology** enabling the system to be developed in iterative phases with continuous testing, feedback and improvements. Each sprint will focus on implementing and evaluating a specific module, such as the dashboard, AI prediction model, inventory management and donation platform.

Tools and Technologies

Frontend

- React + Vite (Fast development and optimized builds)
- Shadcn/UI (Modern, reusable UI components)
- Tailwind CSS (Utility-first styling, required by Shadcn/UI)

Backend

- FastAPI (Python)

- High-performance REST API framework
- Easy integration with AI/ML models
- Automatic API documentation (Swagger/OpenAPI)

Database

- MongoDB

AI & Data Processing

- Python
- Pandas
- NumPy
- Scikit-learn
- Prophet (Time-series demand forecasting)
- Matplotlib

Research Techniques

The following research techniques will be used throughout the project:

- **Literature Review** to study existing food waste management systems and AI-based demand forecasting methods.
- **Requirement Gathering** through surveys and interviews with restaurants, hotels, supermarkets and charitable organizations.
- **Dataset Collection** by obtaining historical sales, inventory and food waste records from food businesses or publicly available datasets.
- **Data Preprocessing** to clean, transform and prepare data for machine learning.
- **Machine Learning Model Training** using historical sales data to predict future food demand.

Methodology Flow

Requirement Analysis

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Data Collection

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Data Preprocessing

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AI Model Training

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Demand Prediction

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System Development

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Testing & Evaluation

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Deployment & Documentation

10. Expected Outcomes

1. **A fully functional AI-powered Food Waste Management System** that enables restaurants, hotels and supermarkets to manage food production and reduce waste efficiently.

2. **An AI-based food demand prediction model** trained using historical sales data to provide accurate recommendations on the quantity of food to prepare each day.

3. **A centralized dashboard** that allows users to monitor sales, inventory, food waste, donations and AI-generated insights in real time.

4. **An inventory and expiry management module** that helps businesses stock levels and receive alerts before food items expire.

5. **A food donation management platform** that connects businesses with nearby charities, enabling surplus food to be donated instead of discarded.

6. **Comprehensive reports and analytics** that provide insights into sales trends, food waste, inventory usage and donation activities, supporting better business decision-making.

7. **A scalable and sustainable solution** that reduces operational costs, minimizes food waste, promotes environmental sustainability and contributes to addressing food insecurity within local communities.

11. Timeline

Phase	Activity	Start Date	End Date
1. Planning	Introduction & Problem Statement, Development Objectives, Software Development Lifecycle Study, Proposal Preparation & Defense	May 20	June 17
2. Design	UI/UX Design, System Analysis, SRS Preparation & Submission	June 22	July 8
3. Implementation	System Development, Module Integration, Core Feature Development	July 9	August 5
4. Testing	Progress Demonstration, Bug Fixing, System Testing, Validation	July 22	August 19

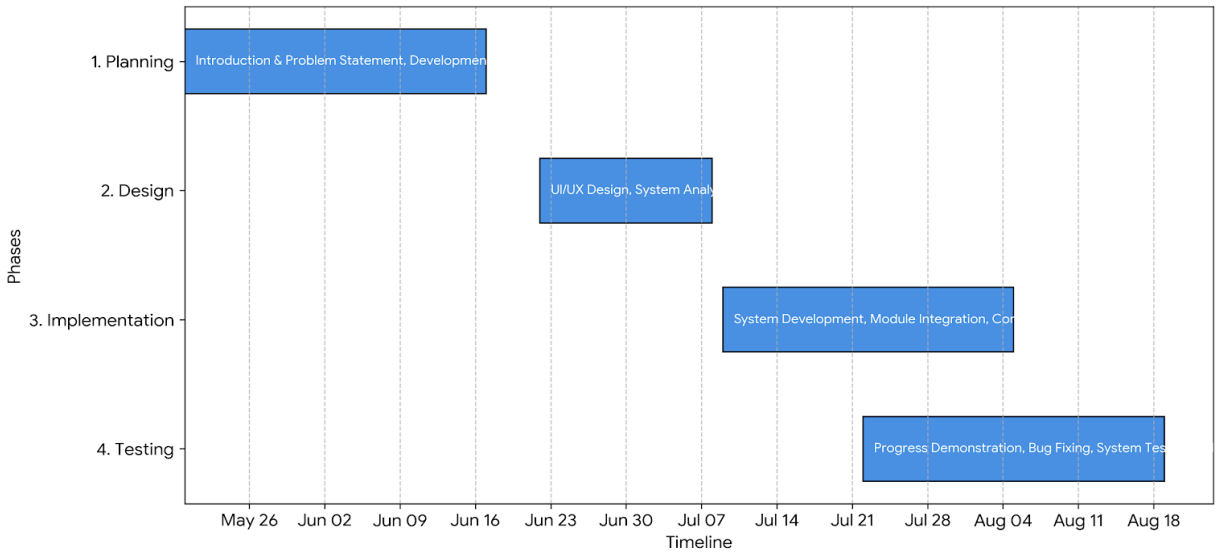


Figure1: Gantt chart

12. Required Resources

Software

- Visual Studio Code (Code Editor)
- Python 3.x
- MongoDB Community Server
- Git & GitHub
- Postman (API Testing)
- Figma (UI/UX Design)
- Google Chrome / Microsoft Edge

Programming Languages

- Python
- JavaScript
- HTML5
- CSS3

Frameworks & Libraries

Frontend

- React
- Vite
- Shadcn/UI
- Tailwind CSS

Backend

- FastAPI

AI & Data Processing

- Pandas
- NumPy
- Prophet

- Scikit-learn
- Matplotlib

Database

- MongoDB

APIs & External Services

- Google Maps API (Location-based Charity Mapping)
- Firebase Cloud Messaging (Push Notifications)
- Email API (Email Notifications)

Dataset Resources

- Historical Sales Data
- Inventory Data
- Food Waste Records
- Donation Records
- Public Holiday Calendar (for improving demand forecasting)
- Public Food Demand Datasets

Documentation & Project Management

- Microsoft Word
- Google Drive (File Sharing)
- Jira (Task Management)

13. References

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14. Signatures

Group Leader: _____ **Date:** _____2026/07/18_____

(Venuja Dodanwala)